

## Question

Write a C program that displays a menu and uses a **switch statement** to print the selected **Increasing Pattern**. Use **nested loops without arrays**. Each pattern should be implemented using a separate **parameterless function with void return type**.

Menu:

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

## Algorithm

1. Start.
2. Declare the required variables.
3. Declare five parameterless functions with void return type:
  - o pattern1()
  - o pattern2()
  - o pattern3()
  - o pattern4()
  - o pattern5()
4. Display the menu.
5. Input the user's choice.
6. Use a switch statement to perform the selected operation.
7. If choice is **1**, call pattern1().
  - o Input number of rows n.
  - o Use nested loops to print i asterisks in row i.
8. If choice is **2**, call pattern2().
  - o Input n.
  - o Use nested loops to print the row number i, i times.
9. If choice is **3**, call pattern3().
  - o Input n.
  - o Use nested loops to print numbers from 1 to i in row i.
10. If choice is **4**, call pattern4().
  - o Input n.
  - o Set num = 1.
  - o Print i consecutive numbers in row i, increasing num after every print.
11. If choice is **5**, call pattern5().
  - o Input n.
  - o Check that n is between 1 and 26.
  - o Use nested loops to print letters from A up to the corresponding letter in each row.
12. If choice is **6**, exit the program.
13. For any other choice, display **Invalid choice**.
14. Repeat the menu until the user selects Exit.

15. Stop.

## Test-Case Table

| Test Case | Choice | Inputs | Expected Outputs                                    | Results |
|-----------|--------|--------|-----------------------------------------------------|---------|
| 1         | 1      | 4      | Increasing * triangle with 4 rows                   | Pass    |
| 2         | 2      | 4      | Rows: 1, 2 2, 3 3 3, 4 4 4 4                        | Pass    |
| 3         | 3      | 4      | Natural numbers increasing from 1 in each row       | Pass    |
| 4         | 4      | 4      | Consecutive numbers from 1 to 10 in triangular form | Pass    |
| 5         | 5      | 4      | Rows from A to A B C D                              | Pass    |
| 6         | 5      | 26     | Alphabet pattern up to Z                            | Pass    |
| 7         | 5      | 27     | Invalid number of rows                              | Pass    |
| 8         | 7      | -      | Invalid choice                                      | Pass    |
| 9         | 6      | -      | Program exits                                       | Pass    |

## C Program

```
#include <stdio.h>
```

```
void pattern1(void);  
void pattern2(void);  
void pattern3(void);  
void pattern4(void);  
void pattern5(void);
```

```
int main()  
{  
    int choice;  
  
    do  
    {  
        printf("\n--- Increasing Patterns Menu ---\n");  
        printf("1. Increasing right-angled triangle\n");  
        printf("2. Increasing row-number triangle\n");  
        printf("3. Increasing natural-number pattern\n");
```

```

printf("4. Consecutive-number triangle\n");
printf("5. Increasing alphabet pattern\n");
printf("6. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        pattern1();
        break;

    case 2:
        pattern2();
        break;

    case 3:
        pattern3();
        break;

    case 4:
        pattern4();
        break;

    case 5:
        pattern5();
        break;

    case 6:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice.\n");
}

} while(choice != 6);

return 0;
}

/* Increasing right-angled triangle */
void pattern1(void)
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

```

```

if(n <= 0)
{
    printf("Invalid number of rows.\n");
    return;
}

for(i = 1; i <= n; i++)
{
    for(j = 1; j <= i; j++)
    {
        printf("* ");
    }
    printf("\n");
}
}

```

```

/* Increasing row-number triangle */
void pattern2(void)
{

```

```

    int n, i, j;

```

```

    printf("Enter number of rows: ");
    scanf("%d", &n);

```

```

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

```

```

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

```

```

/* Increasing natural-number pattern */
void pattern3(void)
{

```

```

    int n, i, j;

```

```

    printf("Enter number of rows: ");
    scanf("%d", &n);

```

```

    if(n <= 0)
    {

```

```

        printf("Invalid number of rows.\n");
        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

/* Consecutive-number triangle */
void pattern4(void)
{
    int n, i, j, num = 1;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }
}

/* Increasing alphabet pattern */
void pattern5(void)
{
    int n, i, j;

    printf("Enter number of rows (1-26): ");
    scanf("%d", &n);

    if(n < 1 || n > 26)
    {
        printf("Invalid number of rows.\n");
    }
}

```

```

        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%c ", 'A' + j - 1);
        }
        printf("\n");
    }
}

```

## Compilation and Execution

```

(kali㉿kali)-[~]
└─$ vi hdoc24.c

(kali㉿kali)-[~]
└─$ gcc hdoc24.c -o hdoc24

(kali㉿kali)-[~]
└─$ ./hdoc24

— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 1
Enter number of rows: 4
*
* *
* * *
* * * *

— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern

```

```
6. Exit
Enter your choice: 2
Enter number of rows: 4
1
2 2
3 3 3
4 4 4 4
```

```
— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 3
Enter number of rows: 4
1
1 2
1 2 3
1 2 3 4
```

```
— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
```

```
Enter your choice: 4
Enter number of rows: 4
1
2 3
4 5 6
7 8 9 10
```

```
— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 5
Enter number of rows (1-26): 4
A
A B
A B C
A B C D
```

```
— Increasing Patterns Menu —
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter your choice: 5
```

```
Enter number of rows (1-26): 26
A
A B
A B C
A B C D
A B C D E
A B C D E F
A B C D E F G
A B C D E F G H
A B C D E F G H I
A B C D E F G H I J
A B C D E F G H I J K
A B C D E F G H I J K L
A B C D E F G H I J K L M
A B C D E F G H I J K L M N
A B C D E F G H I J K L M N O
A B C D E F G H I J K L M N O P
A B C D E F G H I J K L M N O P Q
A B C D E F G H I J K L M N O P Q R
A B C D E F G H I J K L M N O P Q R S
A B C D E F G H I J K L M N O P Q R S T
A B C D E F G H I J K L M N O P Q R S T U
A B C D E F G H I J K L M N O P Q R S T U V
A B C D E F G H I J K L M N O P Q R S T U V W
A B C D E F G H I J K L M N O P Q R S T U V W X
A B C D E F G H I J K L M N O P Q R S T U V W X Y
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

— Increasing Patterns Menu —
```

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

Enter your choice: 5

Enter number of rows (1-26): 27

Invalid number of rows.

— Increasing Patterns Menu —

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

Enter your choice: 7

Invalid choice.

— Increasing Patterns Menu —

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit

Enter your choice: 6

Exiting program...